

BOTANY XII

10 Year Questions Distribution

According to the Chapters.

This Distribution is carried out According to
NEW BOOK OF BIOLOGY.

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QUESTIONS		YEARS
Chapter: 22	CHROMOSOME AND DNA	
Section B		
i) Give the name of enzymes involved in replication of DNA, also give function of any one of them. ii) Differentiate between protein synthesis in Prokaryotes and Eukaryotes. viii) What is heteroploidy? Write its types.		2025
xi) Write a note on any one of the following: * Sickle Cell Anemia * Phenylketonuria		2024
(i) What do you know about sickle cell Anemia OR Alkaptonuria? (iii) State Hershey and Chase experiment that confirmed DNA as a hereditary material. (vi) What are the effects of Chemical Mutagens on DNA?		2023
2. ii) Name various Nitrogen bases of DNA. 2. iii) Differentiate any one of the following: (a) DNA and RNA (b) Autosome and Sex Chromosome 3. ii) Draw a chart to show the chemical composition of chromosome.		2022
2. iii) Mention number of autosomes and sex chromosomes in a human cell 2. vii) Name various types of RNA involved in protein biosynthesis. 3. iv) Mention different types of chromosomes according to the position of centrometre.		2021

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3.vi) Define mutation. How does ionization radiation damage the DNA?	
2. (vi) Draw Karyotype of Drosophila. 2. (xiii) Name various nitrogenous bases found in DNA nucleotides. 3. (iii) Classify chromosome on the basis of position of centromere.	2019
3.iii) What is an autosome? How many Autosomes are there in a human cell?	2018
2.iii) Name the compounds of DNA. 2.iv) Name the scientist known for the following discoveries. (a) ladder type of organization of DNA (b) One gene-one enzyme hypothesis 3. viii) Write note on genetic code. 3. vii) Define merism and mention its types OR Differentiate between DNA and R.N.A.	2017
2. i) Differentiate any One of the following: a) DNA and RNA b) Autosome and Sex-chromosomes 2. viii) Write a note Phenylketonuria Or Alkaptonuria. 2. xiii) Write a note on genetic code.	2016
iii) Explain Chromosomal Aberration. vi) Describe the types of RNA.	2015
2. ii) Differentiate between DNA and RNA. 2. xi) State four point of Chromosomal theory of heredity. 2. xii) Write a note on the significance of Secondary growth.	2014
Section C	
3. Give the evidence of DNA as hereditary material.	2025
4. Explain the term DNA replication. Describe the process of DNA replication with diagram.	2024
4. —OR— Describe the Law of Independent Assortment with the help of genetic cross.	
4. Describe the ultra-structure of chromosome with their chemical composition. Also discuss its types.	2023
6. a) Describe Watson and Crick model of DNA.	2022
7. Give a brief description of structure of DNA.	2021
4. Describe Watson and Crick model of DNA. How does the model explain semi-conservative replication of DNA?	2019
4) Prove with the help of experiment that DNA is a genetic material.	2017

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4 OR] Explain Watson & crick model of DNA and write its replication.	
4. Define Chromosome. Describe with diagram the chemical composition of Chromosome and their ultra-structure.	2016
3. Describe Watson and Cricks Model of DNA. How does DNA replicates semi-conservatively.	2015
Chapter: 23	INHERITANCE
Section B	
iii) Why human males are heterogametic? x) Why Y-linked traits are called holandric Traits? ix) Explain any one of the following: *co-dominance *incomplete dominance	2025
x) Who O-negative blood group is called the universal donor? xii) Write a note on any one of the following: * Incomplete Dominance * Co-Dominance * Test Cross xiii) Why haemophilia is common in male?	2024
2. iv) What is a gene?	2022
3. i) What is Test Cross? Why is it performed?	
3.i) State the law of segregation?	2021
3.iii) Make a cross between a color blind male and normal female and mention result.	
2. (xiv) Which cross helps to determine the homozygosity of a dominant parent?	2019
2.x) Define test cross. 3.ii) Find the possible blood group of children if the blood group of father is O and that the mother AB.	2018
2.vi) Define Allele. 2.x) What will be the result of a cross between Haemophilia- male and Normal-female 2.xii) Give definition of Gene pool OR Crossing over. 2.xiv) State the law of independent Assortment. 3. vi) Give the Cross between Red – eyed female and white – eyed male in drosophila.	2017
2. ix) Draw a chart showing Phenotype and Genotype of different Blood groups. 2. xii) A color blind woman is married with a normal man. How many children of this couple are likely to be color blind in F ₂ generation? 2. xv) State the law of Segregation with Phenotypic ratio and Genotypic ratio.	2016

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2. v) Describe Test cross and Incomplete dominance.	2015
2. vii) Why did Mendal choose pea plant as an experimental material.	
2. iii) A color blind women marries a normal man. How many children of this couple are likely to be color blind in F ₂ generation	2014
2.vii) Draw a chart showing seven pairs of contrasting characters observed by Mendel. Also mention which of these characters are dominant or recessive.	
2. ix) What is a test cross? Explain with example.	
Section C	
4. Describe inheritance of two traits (Dihybrid Cross) with the help of checker board.	2025
4.—OR—Describe multiple allele with the help of ABO blood group.	
4. (OR) State the law of independent assortment. Explain with the help of a checker board.	2023
4. a) Explain multiple allele with the help of Human Blood Group.	2022
5. a) Describe sex linked inheritance in man with reference to color blindness.	
4. State Mendel's law of independent assortment with the help of checker board.	2021
5. Define Sex-linked inheritance. Explain with reference to haemophilia in man	2019
[6 OR] Describe the Law of Independent Assortment with the help of checker board.	2018
5. Describe the 'Law of Independent Assortment'. Explain it with the help of checker board method..	2016
[OR 5] Define Sex-linked inheritance. Discuss Sex-linked inheritance in humans.	2015
3. State the Law of Independent Assortment. Explain it with the help of checker board method.	2014

Chapter: 24

EVOLUTION

Section B

iv) Define Homologous organs with example.	
v) Describe the Invagination theory.	2025
xi) Differentiate between Divergent and Convergent evolution.	
iii) Define Speciation. Name the types of Speciation.	2024

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2(n) vi) Why is Archaeopteryx considered a link between Reptiles and Birds?	2023
2. (a) (vii) How does artificial selection play role in the formation of new specie?	2022
2. b) vii) Write any four objections on Darwin's theory of Natural selection.	2021
2.a) (iv) Why is Archaeopteryx considered a link between Reptiles and Birds?	2019
2.a) (iii) Why is Archaeopteryx considered a connecting link between Reptiles and Birds?	2018
2.b) (xv) State the Objections on Lamarek's Theory.	
2.b) (xiv) State the evidences of vestigial for evolution in man.	2017
(vi) What are Vestigial organs? Give the example with the reference of animals.	2016
(vi) Explain Hardy Weinberg Law.	2015
(iii) Write any four objections on Darwin's Natural Selection Theory.	2014
(xi) Give any two evidences in favour of Organic evolution.	
Section C	
3. Define Evolution. Discuss Lamark's theory and the objection raised against it.	2016
5. Describe Darwin theory of Natural selection and the objections raised against it.	2015

Chapter: 25

MAN AND HIS ENVIRONMENT

Section B

vi) What are Aquifers? What is their role in the Earth's hydrological cycle?	2025
xii) Differentiate between Xerarch and Hydrarch succession.	
xiv) Write a short note on water cycle.	
ii) Describe the causes of ozone layer depletion OR Describe the causes of acid rain.	2024
ix) What are the advantages of nuclear power?	
ix) OR—Name two nuclear power plants of Pakistan.	
xiv) Describe the renewable and non- renewable environmental resources.	
2(a) vii) Why is algal bloom harmful for aquatic animals?	2023

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2. (a) (i) Why is depletion of ozone layer harmful?	2022
2. (b) (xiii) Write note on 'Algal bloom'.	
2.a) (v) How is Ozone layer being depleted?	2019
2.b) (xvi) Write note on Acid Rain OR Depletion of Ozone layer	2018
2.b) (xvi) How is Ozone layer being destroyed?	2017
(ii) Write a note on Lithotrispy or Acid Rain. (iv) Define renewable and non-renewable resources and examples.	2015
Section C	
5. What is meant by Biogeochemical Cycle? Describe Nitrogen cycle.	2025
5. What is meant by Biogeochemical Cycle? Describe Nitrogen cycle.	2024

Chapter: 26	BIOTECHNOLOGY
Section B	
vii) Why restriction enzymes are called molecular scissors? xiii) Define any two of the following: *Primary cell culture *3D culture *Cell line culture *Anther culture	2025
i) RFLPs and GMO are the abbreviations of terms iv) Define any two of the following terms: * Genome * Gel Electrophoresis * Tissue Culture * Genomic Library v) Describe the goal of Human Genome Project (HGP). vi) What is gene therapy? How many methods are there for gene therapy? Write their names. vii) Why restriction enzymes are called 'Molecular Scissors'?	2024
2(b) xiii) Define DNA fingerprint and how is it performed?	2023
2.a) (vi) Why is DNA fingerprinting performed? 2.b) (xv) State the significance of Genetically engineered bacteria.	2019
2.a) (vi) Why are plasmids used in Recombinant DNA. Technology?	2018
2.a) (vii) Why is DNA Finger Printing performed? 2.b) (xiii) Write note on Transgenic animals.	
(xv) Write a note on Human Genome Project.	2017
	2016

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(v) Write the basis steps of recombinant DNA technology.	2015
(x) Write a note on DNA fingerprinting.	
(xv) Write a note on DNA fingerprinting.	2014
Section C	
6. Explain techniques of Gene Cloning by Recombinant DNA technology.	2025
3. What is PCR? Describe gene amplification through PCR. Writes its two applications.	2024
4. Explain various steps involved in recombinant DNA technology. Draw labelled diagram of different steps of insulin formation by this technology.	2023
5. Explain the various steps involved in Recombinant DNA technology and draw neat and labelled diagram	2019
[3 OR] Differentiate between Unifactorial and Multifactorial defects.	2018
5. Describe the different step in DNA recombinant Technology. Draw labelled diagrams of different steps.	2017
[3. OR] Write the various steps involved in recombinant DNA technology with the help of diagram.	2016
5. Write is Gene therapy?	2014

Chapter: 27	BIOLOGY AND HUMAN WELFARE
Section B	
viii) What is meant by home gardening? Write its significance. viii)—OR—Write the names of five vegetables and five fruits of summer season.	2024
Section C	
3.—OR— Write the role of microbes in Food Processing and Sewage Treatment.	2024
5.—OR—What is vaccination? Explain its role in preventing 'Hepatitis' and 'Tetanus'.	